Talha Laique

Machine Learning Engineer | Computer Vision
+47 929 18125| talhalaique9@gmail.com | linkedin.com/in/talha-laique | github.com/laiquet
Ås, Norway | Eligible to work in Norway (Residence Permit – no sponsorship required)
Portfolio: https://laiquet.github.io/

PROFILE

Machine Learning / Computer Vision Engineer with 5+ years of experience building and deploying computer vision, classification, and document-AI systems in production and applied-research settings. Hands-on with the full ML lifecycle: dataset creation and annotation, model training (CNNs, YOLO, SAM), evaluation, and deployment via Docker/Flask/REST APIs with MLflow experiment tracking. Comfortable owning a pipeline end-to-end, from raw image/document data to a shipped, monitored inference service.

TECHNICAL SKILLS

Languages / Core: Python, SQL, NumPy, pandas, Matplotlib
Machine Learning / Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs, YOLO (v8/v11/v26), SAM, Ensemble Learning, Model Evaluat
Computer Vision / Document AI: Image Classification, Object Detection, Instance Segmentation, Image Annotation,
OCR / Document Understanding
MLOps / Deployment: Docker, MLflow, Gunicorn, FastAPI, Flask, REST APIs, Git, Linux, GCP, Azure
LLM / GenAI: Hugging Face, Gemini API, LangChain, ChromaDB, Ollama, RAG

EXPERIENCE

NMBU | Computer Vision Researcher (PhD Candidate) — Ås, Norway | Jan 2024 – Feb 2026

- Built end-to-end computer vision pipelines (detection, segmentation, feature extraction) to quantify melanin-based pigmentation in Atlantic salmon, converting raw images into structured, analysis-ready datasets for a multi-institution welfare study.
- Created and validated annotated image datasets for ROI segmentation and object detection, covering preprocessing, label QA, and structured export for training.
- Delivered reproducible batch-inference and CSV-generation workflows enabling per-fish, per-day, and per-side temporal analysis, supporting two peer-reviewed publications.

INIT ML | Machine Learning Engineer Intern — Paris, France | May 2021 – Oct 2021

- Built an end-to-end product taxonomy model classifying fashion items into 2 category levels and multiple attributes from images and metadata, deployed via a web app and API demo.
- Built scraping and preprocessing pipelines that generated a custom multi-thousand-image e-commerce training dataset.
- Quantized and integrated the trained model into an Android app for on-device product image inference.

BASIKON | Machine Learning Engineer Intern – Document AI — Paris, France | Apr 2020 – Aug 2020

- Built a document-understanding pipeline extracting structured fields from scanned financial documents across multiple layouts and scan qualities.
- Created an in-house annotated dataset of financial document images and prototyped CNN-based field-extraction models.

GOJOB | Data Scientist Intern — Paris, France | Sep 2020 – Mar 2021

- Built ensemble and embedding-based models classifying user career pathways into 17 broad and 57 specific job categories from work-history and education data.
- Engineered pathway features from work experience, education, and word embeddings to improve classification accuracy across both taxonomy levels.

PROJECTS

YOLOV8 SPOT DETECTOR - MLOPS PIPELINE — Python, Flask, YOLOv8, MLflow, Docker, Gunicorn, REST API

- Deployed a Dockerized Flask API serving a custom YOLOv8 model for fish spot detection (86.5% mAP@50), with MLflow tracking for experiment versioning.
- Built a production-style inference pipeline with authentication, rate limiting, and a drag-and-drop web UI for real-time bounding-box visualization.

SALMON YOLO PIPELINE - UNDERWATER FISH DETECTION Python, YOLO26x, Ultralytics, OpenCV, Real-ESRGAN, CUDA, CLI

- Built a 4-stage CLI pipeline (preprocess → train → infer → crop) unifying underwater image enhancement, YOLO detection, and ROI extraction on a 207-image, 1,750-instance dataset.
- Designed an underwater preprocessing pipeline (Gray-World white balance, CLAHE, Real-ESRGAN) improving UIQM by +80% and UCIQE by +176%.
- Fine-tuned YOLO26, reaching 0.756 mAP@50 (+37% over the raw-image baseline).

AI-ASSISTED COMPUTER VISION ANNOTATION TOOL Python, PySide6, PyTorch, OpenCV, SAM, YOLO, COCO

- Built a desktop annotation tool (polygons, bounding boxes, OBBs, undo/redo, autosave, crash recovery) with SAM-prompted and YOLO auto-detection labeling assistance.
- Implemented import/export for YOLO, COCO, JSON, and SAM RLE formats to interoperate across annotation, training, and inference workflows.

AI DOCUMENT OCR & TABLE EXTRACTION JavaScript, Node.js, Express, Gemini 2.5 Flash, Multimodal AI

- Built a full-stack app using Gemini 2.5 Flash to extract structured tables from multi-page financial-document PDFs, with automatic document classification.
- Built an editable results interface with inline cell correction and multi-format export for production-ready data.

SWARM PAPER ANALYZER - MULTI-AGENT RESEARCH INTELLIGENCE Python, LangGraph, LangChain, ChromaDB, Gemini API, RAG

- Architected a 6-agent LangGraph swarm (Supervisor + 5 specialists) with a fault-tolerant arXiv ingestion pipeline and ChromaDB RAG store for semantic retrieval.

EDUCATION

PhD - Computer Vision for Atlantic Salmon Welfare Ås, Norway | Jan 2024 – Present
NMBU
Coursework: Computer Vision, Image Analysis, Animal Stress Biology

MScT - Artificial Intelligence and Advanced Visual Computing Paris-Saclay, France | Sept 2021
Université Paris-Saclay – 3.67/4.0
Coursework: Computer Vision, Deep Learning, Machine Learning, NLP, Reinforcement Learning

BSc - Computer Engineering Topi, Swabi, Pakistan | June 2015
GIK Institute – 3.22/4.0
Coursework: Data Mining, Databases, Data Structures & Algorithms, Digital Image Processing

PUBLICATIONS

- Lewis, R., Kostermans, T., Brovold, J. W., **Laique, T.**, and Ocepek, M. *Automated Body Condition Scoring in Dairy Cows Using 2D Imaging and Deep Learning.* AgriEngineering, 7(7), 241, 2025.
- **Laique, T.**, Gunnes, M., Folkedal, O., Nilsson, J., Green, E. A. L., Gundersen, H. N., Øverli, Ø., and Ullah, H. *Quantification of Opercular Pigmentation Changes in Farmed Atlantic Salmon.* Fishes, 11(5), 271, 2026.
- Ijaz, A. Z., Ali, R. H., Ali, N., **Laique, T.**, and Khan, T. A. *Solving Graph Coloring Problem via Graph Neural Network (GNN).* ICET 2022, pp. 178–183.